Exercise 2

Fix positive integer K and consider the K-nearest neighbor regression with $\mathcal{X}$ being any set (any metric space) and $\mathcal{Y} = \mathbb{R}$ who gives a fitted model

$$\hat{f}(x; \mathcal{D}) = \frac{1}{K} \sum_{i \in N(x, \mathcal{D})} y_i,$$

where $\mathcal{D}$ is any data set containing points from $\mathcal{X} \times \mathcal{Y}$ and $N(x, \mathcal{D})$ denotes indices of all K points in $\mathcal{D}$ whose $\mathcal{X}$ component is, in some sense, closest to x. We can assume that there is almost surely no ties by implementing any tie breaking rule.

Now let us consider the data set

$$\mathcal{D}_n = \{(X_1, Y_1), \cdots, (X_n, Y_n)\}.$$

For any $y \in \mathcal{Y}$, put

$$\mathcal{D}_{n+1}^y = \{(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, y)\}.$$

Prove that there exists real valued functions $a(x; \mathcal{D}_n, X_{n+1})$ and $b(x; \mathcal{D}_n, X_{n+1})$ whose values depend only on x, $\mathcal{D}_n$, and X_{n+1} such that, for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$:

$$\hat{f}(x; \mathcal{D}_{n+1}^y) = a(x; \mathcal{D}_n, X_{n+1}) + b(x; \mathcal{D}_n, X_{n+1}) y.$$

Solution First observe the fact that the set $N(x, \mathcal{D}_{n+1}^y)$ merely reads the set of all indices $i \in \{1, \ldots, n+1\}$ belonging to the K points amongst $X_1, \ldots, X_{n+1}$ who are the closest to x. Therefore, although y is present in the notation, $N(x, \mathcal{D}_{n+1}^y)$ depends only on x and the points $X_1, \ldots, X_{n+1}$. Therefore, by simply writing

$$\hat{f}(x; \mathcal{D}_{n+1}^y) = \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{1, \ldots, n\}} y_i + \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{n+1\}} y_i = \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{1, \ldots, n\}} y_i + \frac{1}{K} \chi_{\left\{n+1 \in N(x, \mathcal{D}_{n+1}^y)\right\}} y,$$

we see that it suffices to define

$$a(x; \mathcal{D}_n, X_{n+1}) = \frac{1}{K} \sum_{i \in N(x, \mathcal{D}_{n+1}^y) \cap \{1, \ldots, n\}} y_i$$

and

$$b(x; \mathcal{D}_n, X_{n+1}) = \frac{1}{K} \chi_{\left\{n+1 \in N(x, \mathcal{D}_{n+1}^y)\right\}}.$$

This completes the proof.////